team that has thoroughly tested the code and will maintain the library.

Likewise, if you need an advanced clustering algorithm, like *DeLiClu*, it will be hard to find even Java or Python implementations that can be trusted to run as part of your application in production.

Another situation in which you might need to understand the internals of these algorithms is when you need to customize one of them, either because you need to optimize it for a real-time environment, because you need some specific property (for example tweaking it to run concurrently and be thread-safe) or even because you need a slightly different behavior.

That said, even focusing on the first part, understanding when and how to use the data structures we present, will be a game changer letting you step up your coding skills to a new level. Let’s use an example to show the importance of algorithms in the real world, and introduce our path in describing algorithms.

1.4 Packing your Knapsack: Data Structures Meet the Real World

Congrats, you have been selected to populate the first Mars colony! Drugstores on Mars are still a bit short of goods… and hard to find. So, you will have, eventually, to grow your